

Spanish Vowel Sound Introduction and *VOCAL-ize* Tutorial

In the following slides, you will learn about the differences between the Spanish and English vowel sounds and why it is important to practice them, specifically.

As a short introduction, vowel sounds in both Spanish and English make up the core of every syllable and word. Pronouncing Spanish consonant sounds in ways that differ from L1 (first language) speakers, perhaps due to the influence of a learner's L1, such as the *t* or *d*, can contribute to what L1 speakers of Spanish may call an accent.

However, pronouncing the vowel sounds differently can cause a change in the meaning of any given word, which may cause L1 speakers to misunderstand what you intended to say. Therefore, explicitly training the vowel sounds is very important for your Spanish communication. Nonetheless, many Spanish classes lack the resources to enable you to practice explicit vowel learning. That is why my team and I created *VOCAL-ize*, an app that shows you speech feature differences between L1 Spanish and English speakers that you can compare to your own speech with several practice sessions for each Spanish vowel [i, e, a, o, u] in different syllables, words, phrases, and sentences.

Watch the videos on the next two slides to understand in more depth the Spanish vowel system according to the 5 main single sound Spanish vowels [i, e, a, o, u].

Before watching the videos make sure to keep in
mind a few things...

- Please keep in mind that when the author of the video content says phrases like “learning vowel sounds to sound more native” or “this pronunciation is problematic”, that this can make you feel discouraged because it may make you feel like you will never sound like a “native speaker”. A native speaker is the same thing as an L1 (first language) speaker. Don’t worry about this as you begin your journey with VOCALize.

- While training vowel accuracy is important for understanding on the part of a listener and on your part as the speaker emitting a message, please remember that you do not need to reach perfection to achieve legitimacy in your language goals.

- Problematic is not a word I would use; however, the author would like to describe communication as problematic when vowel sounds vary to the extent that the meaning of a word changes if an English vowel replaces a different Spanish vowel sound in speech.

- If your goal is to become native sounding, this is a legitimate goal. If your goal is to effectively communicate in your L2 (second language) Spanish so that you may convey your intended message for an L1 speaker to understand, then communicability is also a legitimate goal.

- One more thing to note: the video author uses sad and happy faces to demonstrate an Americanized English-Spanish pronunciation of many Spanish words for each vowel. It's important to know that this will not apply to you in the app, and you shouldn't internalize good or bad about your Spanish. Any Spanish that you are using with the app, in class, or in your community is valued, valid, and further trains your pronunciation and fluency as a whole. Remember to think of any utterance as a happy face that may need practice to be able to communicate what you would like in your Spanish.

Why explicitly learn and practice the Spanish vowel sounds?

(Dreaming Spanish, n.d.-a)

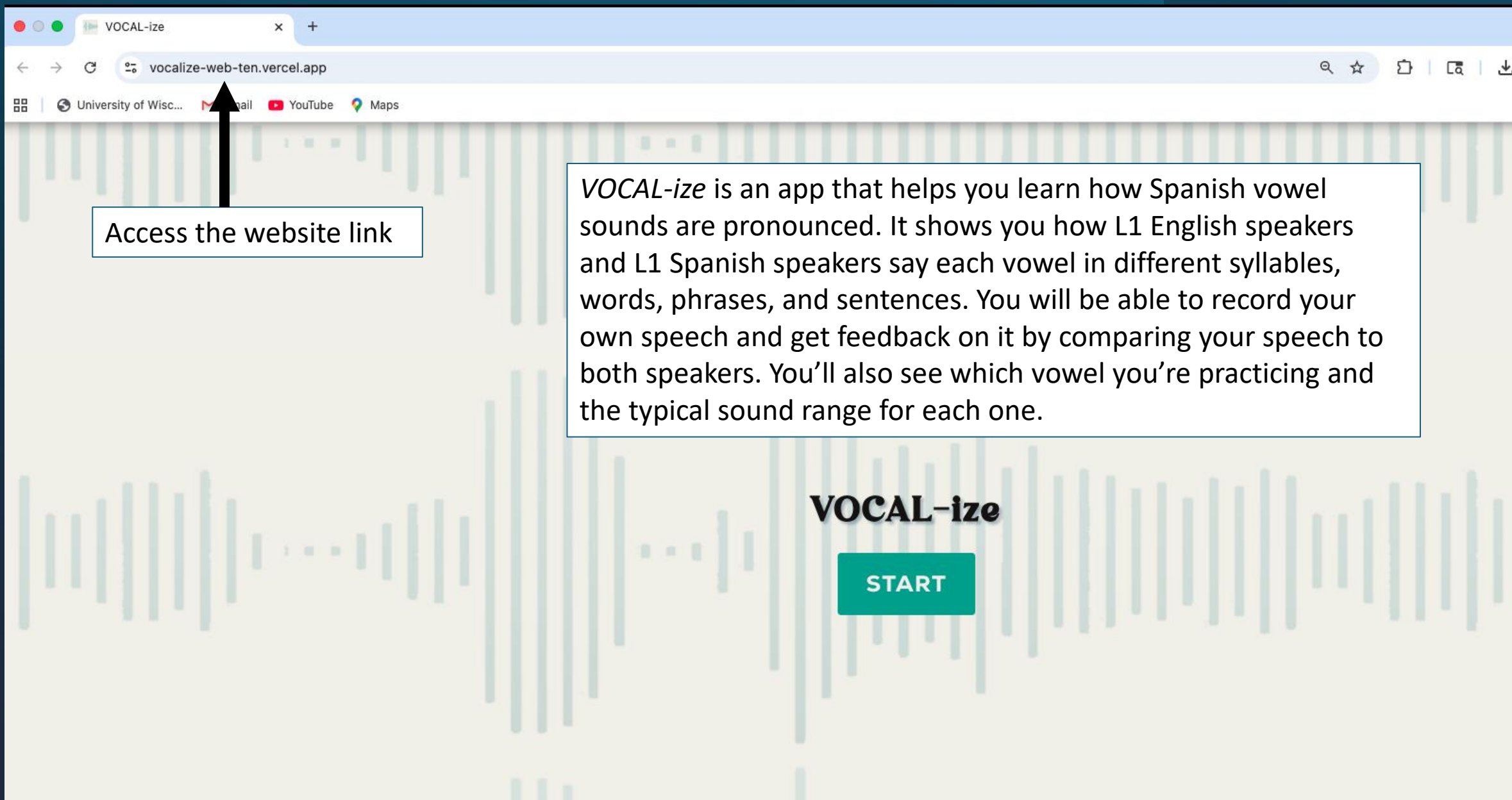


Why explicitly learn and practice the Spanish vowel sounds?

(Dreaming Spanish, n.d.-b)



How do I access the VOCAL-ize app
and what is it?

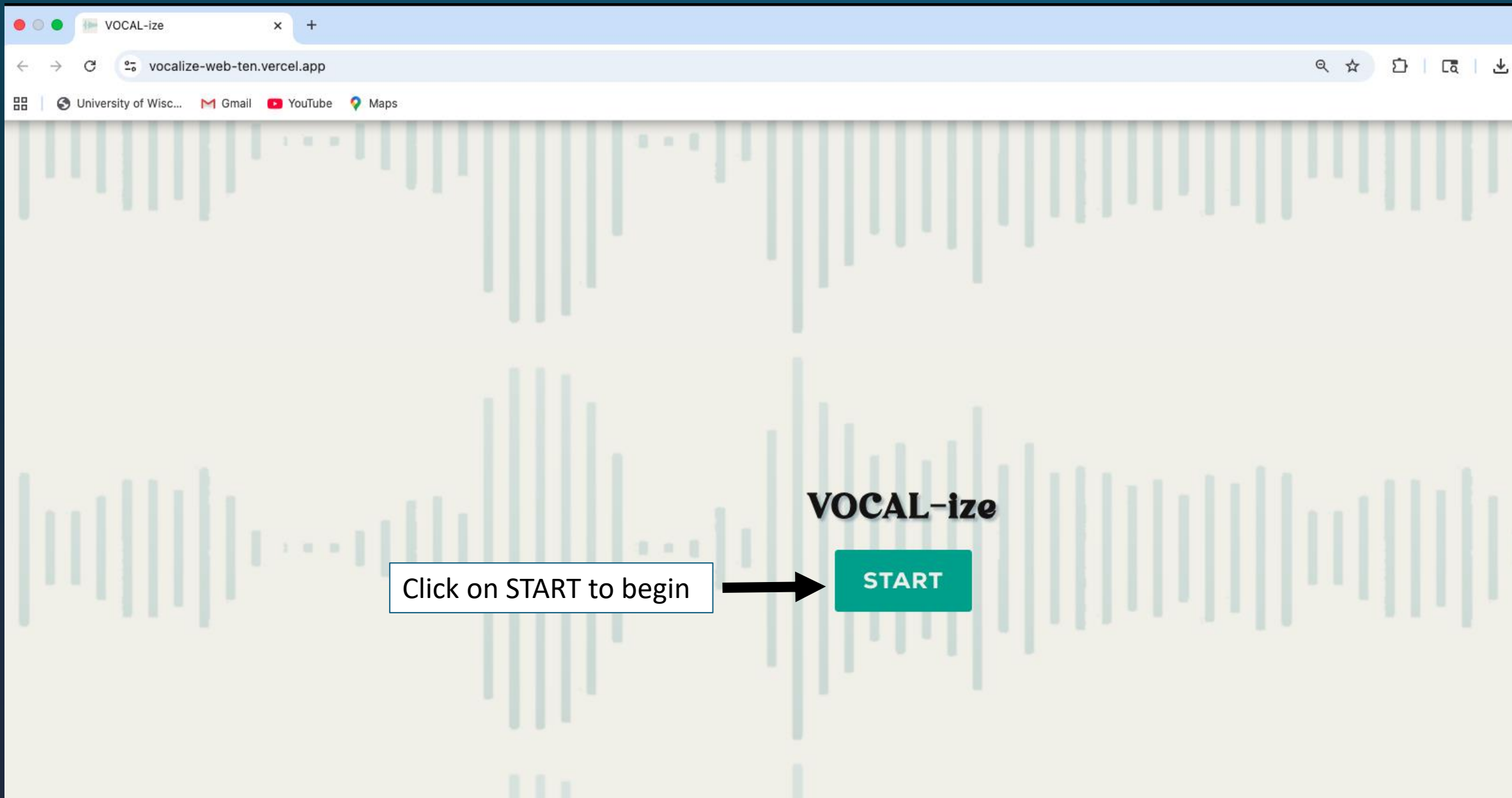


Access the website link

VOCAL-ize is an app that helps you learn how Spanish vowel sounds are pronounced. It shows you how L1 English speakers and L1 Spanish speakers say each vowel in different syllables, words, phrases, and sentences. You will be able to record your own speech and get feedback on it by comparing your speech to both speakers. You'll also see which vowel you're practicing and the typical sound range for each one.

VOCAL-ize

START



What are the two main modules?

There are two different modules you will be working in, *Vowel* and *Stress*.

Modules

In the Stress module, you will practice producing the vowels according to lexical stress, that is, where to put the emphasis on the correct vowel in any given Spanish word (this will be revisited later in the tutorial). You will practice this across the vowel sounds [i, e, a, o, u].



In the Vowel module, you will practice your pronunciation with the 5 main vowel sounds, [i, e, a, o, u]. The '[]' around the vowel sounds indicates that these are sounds instead of the grapheme (letter we read).

Modules

Vowel

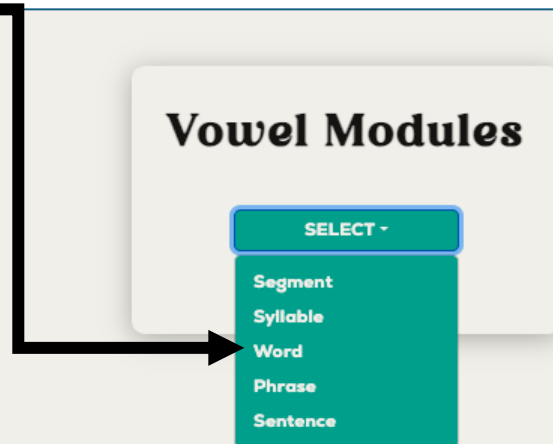
Stress

We will exemplify how the app works by doing a practice in the Vowel module. Click on the module 'Vowel'.

How do I use the Vowel Module to help train my vowel pronunciation?

The Vowel module has the Spanish vowels [i, e, a, o, u] in different levels of speech: segment e.g., *a*, syllable e.g., *ma*, word e.g., *mapa*, phrase, e.g., *mapa grande*, and sentence, e.g., *El mapa es grande*.

Select the tab *Word*.



Practice

Submodule: Word

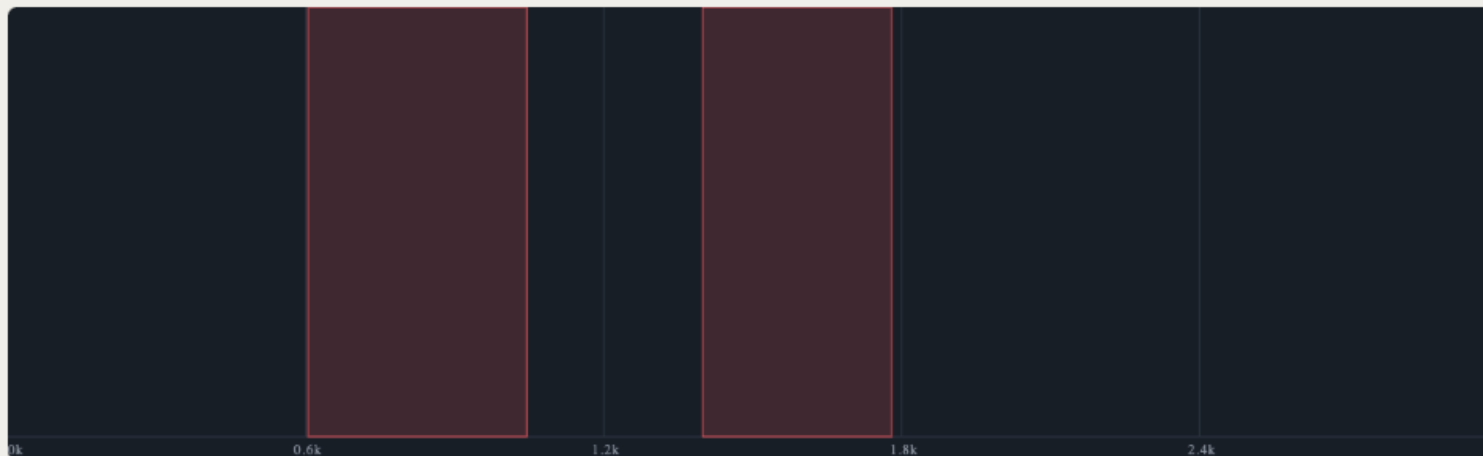
Say: p **a** to

Next

Click play to hear L1 Spanish audio

0:00 / 0:00

[a] v



Start Capture Audio

Advanced Settings

The highlighted context of the word is presented here, the 'a' in *pato*. First, listen to the L1 Spanish audio and pause the audio on the 'a' sound. You can adjust the speed of the audio by clicking on the toggle and clicking 0.5 in the playback speed feature.

Practice

Submodule: Word

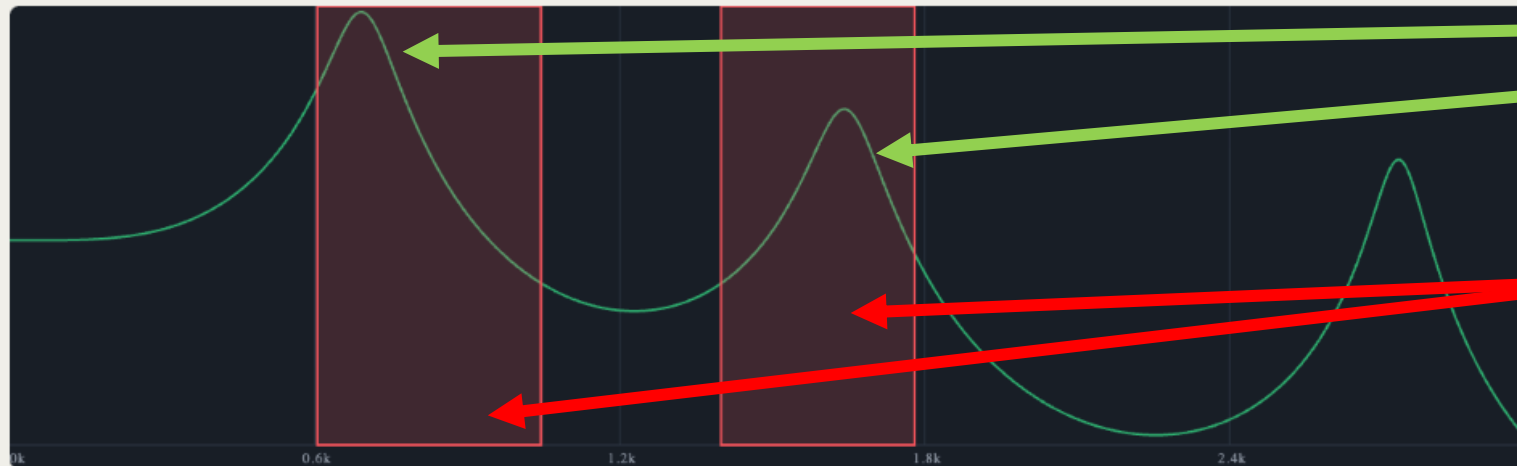
Say: **pato**

Next

Click play to hear L1 Spanish audio

0:00 / 0:00

[a] v



Start Capture Audio

Advanced Settings

You will know if you've paused the audio on the [a] correctly because you should see two to three clear peaks in the spectrogram (visual way to show speech over time) that are green. The green is representing the L1 Spanish speaker.

The **peaks** are called **formants**. Formant values are the resonant frequencies in your voice that show how you shape vowel sounds with your mouth and tongue. The first peak, or the formant (F1), reflects how open your mouth is and how high or low your tongue is, while the second peak, formant (F2), shows how far forward or back your tongue is. By measuring these values, we can see clear patterns for each Spanish vowel. The red squares show the formant ranges of a typical L1 Spanish speaker for the first and second peaks (formants). These peaks help compare how L1 Spanish and L1 English speakers pronounce vowels, showing you how close your own speech is to L1 (native) Spanish pronunciation.

Now go ahead and 'Start Capture Audio' and record yourself saying the target word *pato* above.

Submodule: Word

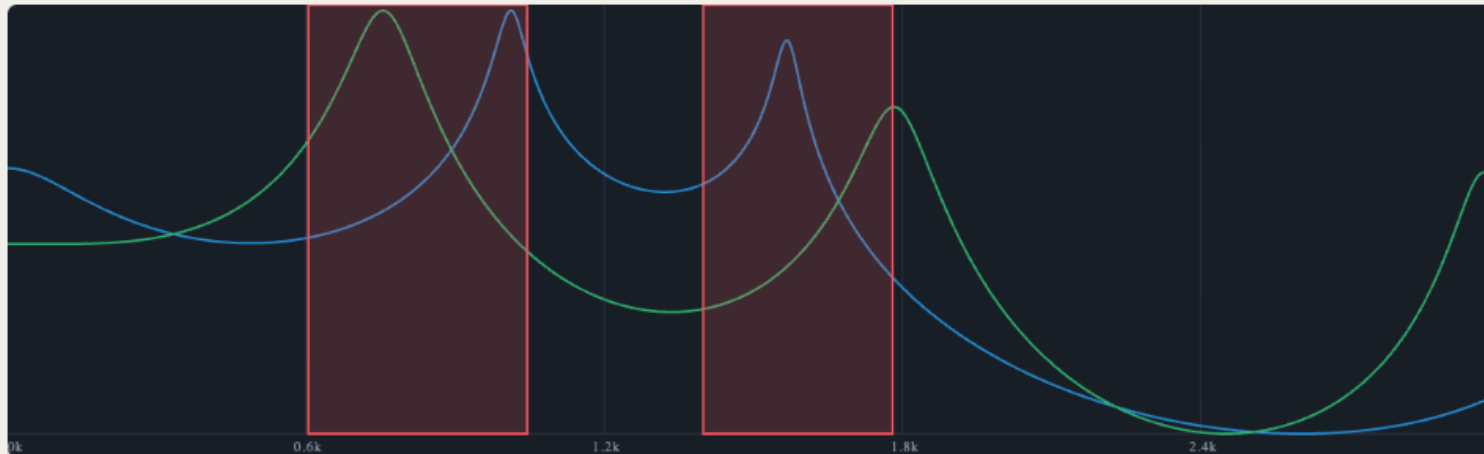
Say: **pato**

Next

Click play to hear L1 Spanish audio

0:00 / 0:00

[a] v



Start Capture Audio

0:00 / 0:01

Playback Speed: 1x v

Playback your recording above.

Advanced Settings

Here is a recording of me saying *pato*, then playing it back at half speed and pausing on the [a] sound after listening to the L1 Spanish audio and pausing it on [a]. While my two blue peaks aren't identical to those of the green L1 Spanish speaker, they fall within the L1 speaker range (red spaces) showing that my [a] pronunciation is close to the target. To keep practicing, click "Next" to explore more target words in any submodule. Each submodule in the Vowel Module includes 10 practice items to help you improve your accuracy. Throughout this semester you will be assigned a specific vowel-module-submodule each week to complete, so don't feel rushed to get through everything at once.

Why is there a stress module? Why is it important for Spanish vowel learning?

In English vowels, we do something called ‘unstressed vowel reduction’ where we reduce any vowel sound to an unstressed vowel, such as schwa, the ‘uh’ sound, in an unstressed context. For example, when we say ‘banana’, the stress goes on the second [a], so we say the other two as ‘uh’, e.g., *buh-na-nuh* (roughly).

In Spanish, the vowels keep a similar sound quality in stressed and unstressed contexts of speech. In Spanish, the second [a] is also the stressed syllable of the word *banana*. A similar [a] quality is maintained for all three [a] sounds in banana, with the second having the emphasis, e.g., *bah-nah-nah* (roughly).

This phenomenon in English can affect your Spanish vowel pronunciation and may again cause challenges in your communication with other Spanish speakers. That is why my team and I included a *Stress* module, allowing you to practice vowel quality in stressed and unstressed contexts.

Watch the video on the following page up to minute 11.45 for a more in-depth discussion of this phenomenon and examples in Spanish vs. English. Remember that the video uses sad and happy faces to express good vs. bad pronunciation. Recall that any Spanish you are using is valid and growing your ability to speak!

Why is there a stress module? Why is it important for Spanish vowel learning?

(Dreaming Spanish, n.d.-c)



How do I use the Stress Module to help train my vowel pronunciation?



Modules

Vowel

Stress

Click on the Stress module.

Stress Modules

SELECT -

Segment

Word

Phrase

Sentence

Click on the Word submodule for an example practice (this same practice will apply to all target practices under the Stress module in this app).

Practice

Submodule: Word

Say: pa to

Next

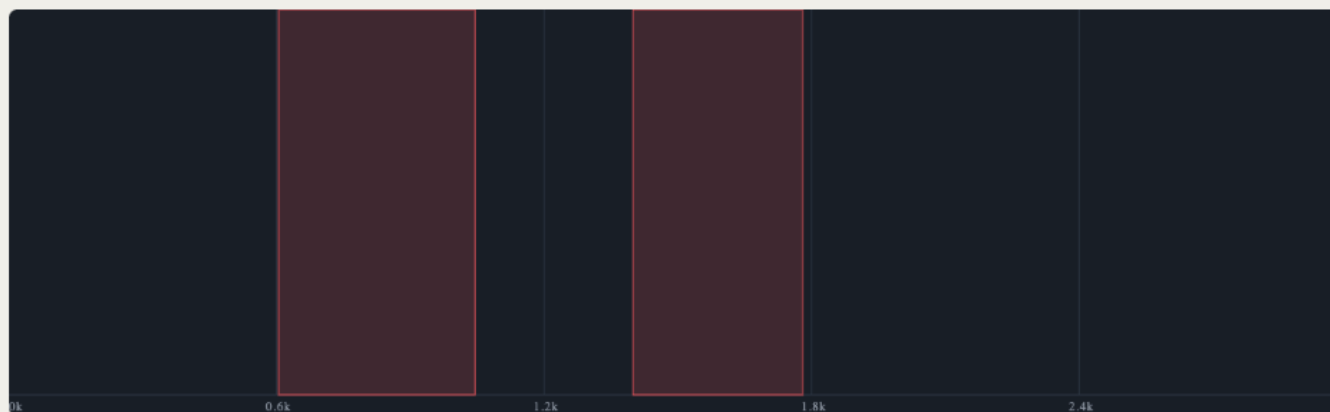
Click play to hear L1 Spanish audio

▶ 0:00 / 0:00

Click play to hear L1 English audio

▶ 0:00 / 0:01

[a] ▼



Start Capture Audio

Advanced Settings

You will listen to and pause each L1 speaker's audio just as you did in the Vowel Module. The difference here is that you will see visual peaks that let you compare the vowel quality in stressed versus unstressed vowels (individually) in Spanish. A list of the stressed and unstressed vowels for each target word, phrase and sentence is provided at the end of this tutorial.

Say: **pato**

Next

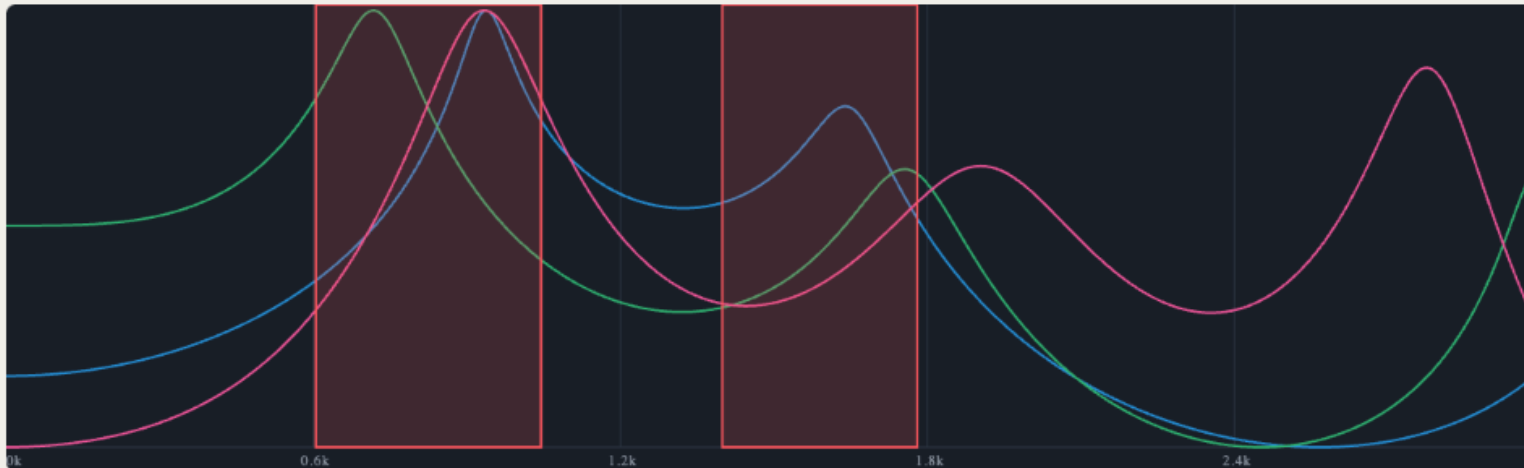
Click play to hear L1 Spanish audio

▶ 0:00 / 0:00

Click play to hear L1 English audio

▶ 0:00 / 0:01

[a] ▼



Start Capture Audio

▶ 0:01 / 0:02

Playback Speed: 1x ▼

Playback your recording above.

Advanced Settings

Here is the word “pato” again, this time for stress. The stressed syllable is the [a] sound. Notice that the second peak for the pink line (L1 English speaker) falls outside the range compared to the green (L1 Spanish speaker) and the blue (me). This means my [a] sound is closer to the target pronunciation than the L1 English speaker’s, which makes sense because of the vowel and reduction differences between English and Spanish. You will do this same type of practice throughout the Stress Module under the Word, Phrase, and Sentence submodules. Each submodule includes 4 stressed and 4 unstressed practice items, and a full list of the corresponding vowels for each target word is provided at the end of this tutorial.

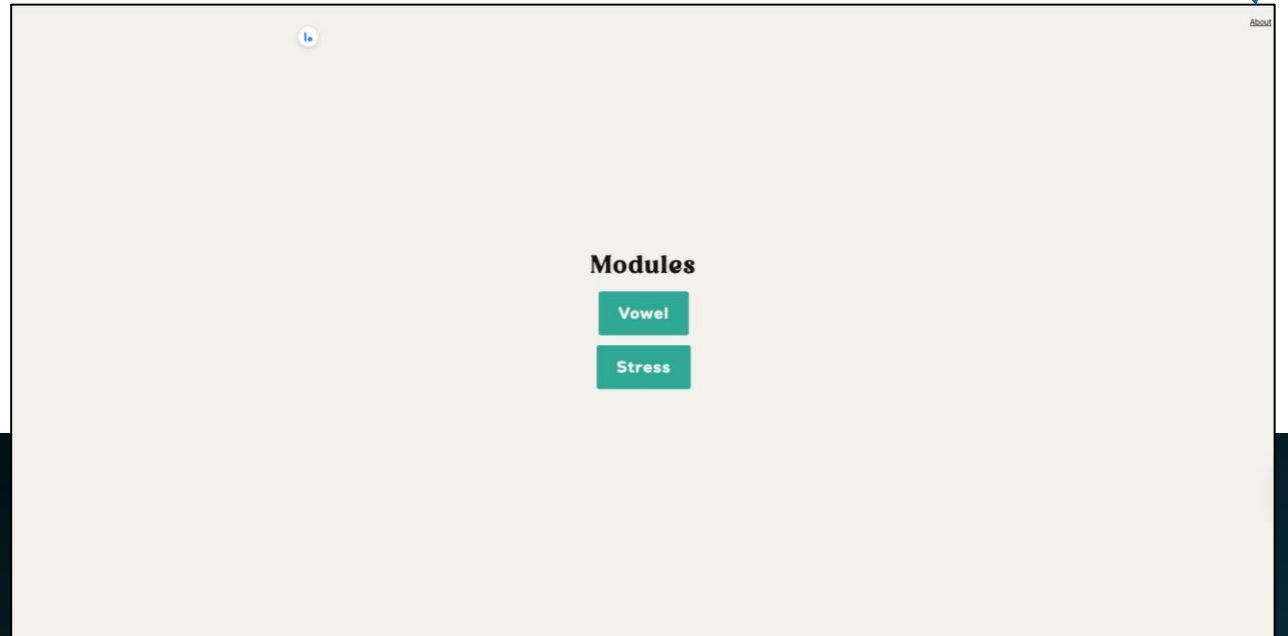
What are the rules for stress (emphasis on a specific vowel) in Spanish?

Watch this video to learn more about the stress rules in Spanish. The list of words in the VOCALize practice are at the end of this tutorial and highlighted, however, this video will give you a more concise picture of assigning stress in Spanish words.

(MaestroKaplan, 2017)



What is the about
page about?



The about page has a short explanation of the app, our team, the app repository (code), an abstract of the larger project I am conducting, and lastly a review and disclaimer of L1 variation in Spanish vowel sounds, which is an acknowledgement of some variation that occurs in the vowel sounds of different Spanish speakers from different countries.

About VOCAL-ize

VOCAL-ize is an educational web app to practice pronunciation with visual feedback. It provides stimuli, recording, and spectral envelope visualization.

Thanks to our volunteers for providing stimulus audio. We are working on adding more stimuli and features, to better capture variation in Spanish and improve usability.

Our team includes Annika Brody Wallander (Project Lead), Frank Jie Zhang (Developer), and Camden Zhu (Developer).

Check out the project repository on GitHub: <https://github.com/timeglitch/VocalizeWeb>
Feel free to contribute or report issues!

► **Summarized Project Abstract**

Back

Stress Module words, phrases and sentences

Vowel	Stress	Word	Phrase	Sentence
[i]	stressed	díva	díva famosa	La famosa díva canta bien.
	stressed	libre	libre paso	Dejan totalmente libre su camino.
	stressed	píco	píco largo	El pájaro muestra un píco largo.
	stressed	tíza	tíza blanca	Toman la tíza blanca.
	unstressed	camínata	camínata larga	Sigo una gran camínata larga.
	unstressed	girasoles	campo de girasoles	Miro siempre ese campo de girasoles.
	unstressed	caríñosa	voz caríñosa	Tiene voz caríñosa.
	unstressed	animal	animal feliz	El pequeño animal corre feliz.
	stressed	pera	pera dulce	Siempre comen una pera dulce.
	stressed	tela	tela suave	Cose esa tela suave.
[e]	stressed	lema	lema natal	Hablan sobre el lema natal.
	stressed	media	media limpia	Lava toda la media limpia.
	unstressed	delicado	tema delicado	Comentan un tema delicado.
	unstressed	necesario	paso necesario	Ese paso técnico es necesario.
	unstressed	pesado	evento pesado	Organiza bien ese evento pesado.
	unstressed	celeste	cielo celeste	Veo de lejos un cielo celeste.
	stressed	malo	niño malo	Dicen que ese niño resulta malo.
	stressed	bata	bata suave	Llevo puesta esa bata suave.
	stressed	pato	pato blanco	Miro un pato blanco.
	stressed	cama	cama nueva	Preparo la cama nueva.
[a]	unstressed	banana	banana madura	Como una banana madura.
	unstressed	palabra	palabra sabia	Dice claramente esa palabra sabia.
	unstressed	caracol	caracol lento	Avanza ese caracol lento.
	unstressed	rodilla	rodilla derecha	Examinan bien la rodilla derecha.

Vowel	Stress	Word	Phrase	Sentence
[o]	stressed	lobo	lobo gris	El lobo gris caza.
	stressed	poco	poco ruido	Hay poco ruido.
	stressed	goma	goma dura	Toca la goma dura.
	stressed	roto	plato roto	Tiro solo ese plato roto.
	unstressed	domingo	día domingo	Salimos el día domingo.
	unstressed	noticia	noticia breve	Recibe cierta noticia breve.
	unstressed	colina	alta colina	Veo desde lejos una alta colina.
	unstressed	botella	botella sucia	Rompo otra botella sucia.
	stressed	mudo	niño mudo	De pronto, ese niño queda mudo.
	stressed	lupa	lupa buena	Tomo la lupa buena.
[u]	stressed	duro	proceso duro	Comenzamos otro proceso duro.
	stressed	rudo	tono rudo	Usa siempre ese tono rudo.
	unstressed	culebra	culebra verde	Vive por aquí la culebra verde.
	unstressed	cuñada	cuñada menor	Abrazamos la cuñada menor.
	unstressed	nublado	día nublado	Camino durante cualquier día nublado.
	unstressed	buzón	buzón de correo	Buscamos el buzón de correo.
Stress Module words, phrases and sentences				

References

- MaestroKaplan. (2017, September 26). *Spanish Accent (pronunciation) rules – Las reglas de acentuación* [Video]. YouTube. <https://www.youtube.com/watch?v=H7HlxXgKill> YouTube
- a. Dreaming Spanish. (n.d.). *Stop reducing vowels when you speak Spanish* [Video]. YouTube. <https://www.youtube.com/watch?v=IDfwlSE6mH0> YouTube
- b. Dreaming Spanish. (n.d.). *Spanish Vowels Part II: [i], [o], and [u]* [Video]. YouTube. <https://www.youtube.com/watch?v=3hDGlg1D6Cs> YouTube
- c. Dreaming Spanish. (n.d.). *Spanish Vowels Part I: Discussion, [a], [e]* [Video]. YouTube. <https://www.youtube.com/watch?v=SY2kunnIWmA> YouTube